Mini Project Phase-I Report On

AI-POWERED SMART ATTENDANCE WEB APPLICATION

Submitted in partial fulfillment for the degree of Bachelor of Technology in

Electronics and Communication Engineering

Submitted by

Manya Naik (40)

Gayatri Mahajan (34)

Under the Guidance of

Prof. Bharat Patil

USHA MITTAL INSTITUTE OF TECHNOLOGY

S.N.D.T WOMEN'S UNIVERSITY

JUHU TARA ROAD, SANTACRUZ (WEST), MUMBAI – 400049

2025–2026

CERTIFICATE

This is to certify that **Manya Naik (40)** and **Gayatri Mahajan (34)** have successfully completed the Mini Project Phase-I report on the topic **"AI-Powered Smart Attendance Web Application"** satisfactorily in partial fulfillment for the Bachelor's Degree in **Electronics and Communication Engineering** under the guidance of **Prof. Bharat Patil** during the year **2025–2026**, as prescribed by Usha Mittal Institute of Technology, Shreemati Nathibai Damodar Thackersey Women's University (SNDTWU).

Guide	Head of Department	Principal
Prof. Bharat Patil	Dr. Shikha Nema	Dr. Yogesh Nerkar

Examiner 1	Examiner 2

DECLARATION

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated any data or facts in this submission.

Manya Naik (40)

Gayatri Mahajan (34)

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our guide, **Prof. Bharat Patil**, for his constant support and valuable suggestions throughout this Mini Project Phase-I. We also thank **Dr. Shikha Nema**, Head of Department, and **Dr. Yogesh Nerkar**, Principal, for providing guidance and necessary facilities. We acknowledge the cooperation of our peers and institute for their encouragement in the completion of this report.

ABSTRACT

The project titled **AI-Powered Smart Attendance Web Application** is designed to overcome the drawbacks of traditional attendance systems that are slow, error-prone, and vulnerable to proxy marking. The proposed system integrates **Face Recognition**, **Wi-Fi Validation**, and **Lecture Token Verification** to ensure secure and accurate attendance tracking.

The system provides a **web-based interface** for students and teachers, supported by a **Python backend** for face detection, network verification, and token validation. By enforcing multiple security layers, it prevents unauthorized or proxy attendance.

This project aims to deliver a **time-efficient, user-friendly, and scalable** attendance management solution that enhances academic integrity.

Keywords: AI Attendance System, Face Recognition, Wi-Fi Validation, Token Authentication, Smart Classroom

Contents

1 INTRODUCTION

1.1 Key Features

- AI-powered Face Recognition for automated attendance.
- Wi-Fi Validation to confirm presence inside the classroom.
- Time-sensitive Lecture Token to prevent replay attacks.
- Teacher Dashboard for monitoring real-time attendance.
- Student Portal for viewing attendance records.

1.2 Background

Traditional roll-calls and biometric systems consume time and are prone to proxy. RFID cards can be misused by others, while Wi-Fi-only systems fail to confirm identity. With the advancement of **AI and computer vision**, face recognition provides a robust method to authenticate presence.

1.3 Purpose

The purpose is to create a **secure, AI-powered, web-based application** that automates attendance, saves time, and ensures fairness by eliminating proxy attendance.

1.4 Problem Statement

Existing attendance systems do not integrate **identity verification and location validation** effectively. There is a lack of cost-effective and proxy-proof systems suitable for educational institutions.

1.5 Objectives of the System

- Automate attendance using AI and web technologies.
- Prevent proxy attendance using a multi-layered security model.
- Provide teacher and student modules with real-time records.
- Enhance reliability and transparency in attendance management.

2 OVERALL DESCRIPTION

2.1 Need of System

Manual roll-call wastes instructional time, proxy attendance reduces fairness, and biometric devices are costly. A **web-based AI solution** ensures efficiency, security, and low cost.

2.2 System Benefits

- Prevents misuse through multi-factor verification.
- Provides transparency for students and teachers.
- Improves institutional integrity.
- Reduces workload for faculty.

3 LITERATURE REVIEW

3.1 Literature Survey

Sr. No	Title	Publish Year	Author/ Org.	Limitation
1.	Face Recognition based Attendance Management System	2020	Smitha, Pavithra S Hegde, Afshin	Low Security, Used LBPH algorithm, proxy marking can possible, lacks real-time integration
2.	Smart Attendance Management System using Face Recognition	2021	K. V. Prasad Reddy, R. Chaitanya Latha, M. Lohitha, R. Sonia, A. B. Usha	Works only in limited environments; does not handle multiple faces or occlusions well

Sr. No	Title	Publish Year	Author/ Org.	Limitation
3.	Smart Attendance System using Face Recognition	2023	Virendra Swaroop Sangtani, Rahul Tak, Rahul Nagarwal, Rinku Yadav	Less robust for live classroom conditions; unable to mark multiple students at once; system crashes when dataset exceeds threshold
4.	Attendance System using Face Recognition	2021	Sarvath Sulthana, K. Shreya, P. Mamatha, K. S. Sushmitha, K. Athmaranjan	requires manual training; recognition accuracy decreases with background variation and head movement.

4 PROPOSED SYSTEM

The **AI-Powered Smart Attendance Web Application** integrates:

1. Face Recognition using AI models (OpenCV + ML).
2. Wi-Fi Validation for classroom-bound authentication.
3. Token Verification for time-specific sessions.

This architecture ensures accuracy, fairness, and security.

5 ARCHITECTURAL OVERVIEW

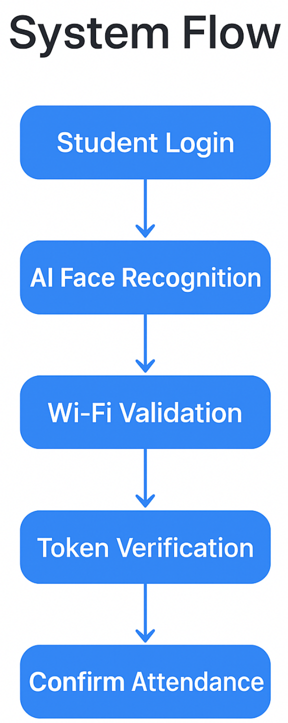
5.1 System Architecture

- Frontend: HTML, CSS, JavaScript (student & teacher portal).
- Backend: Python Flask (AI face recognition + logic).
- Database: MySQL/SQLite for student & attendance records.
- Hardware: Smartphones/laptops with camera + classroom Wi-Fi.

5.2 System Flow

1. Student Login :- Each student logs into the web application using their unique credentials (e.g., roll number, ID, or registered email). This ensures that only authorized users can access the system and begin the attendance process.
2. AI Face Recognition :- After login, the system activates the camera and uses an AI-based face recognition module to verify the student's identity in real time. This step eliminates proxy attendance by matching the live image with the stored face data.
3. Wi-Fi Validation :- Once the face is verified, the system checks the student's connection to the institution's authorized Wi-Fi network. This ensures the student is physically present within the campus premises during attendance marking.
4. Token Verification :- A unique session token (generated per lecture or subject) is validated. This token helps prevent repeated or unauthorized attendance submissions and ensures that attendance is marked only for the ongoing class.

5. Confirm Attendance :- After all validations—login, face, Wi-Fi, and token—the system confirms and records the student's attendance in the database. A success message is displayed to the student and the attendance record is updated for the faculty.



5.3 Frameworks

- OpenCV + ML libraries – Face recognition.
- Flask – Backend server.
- SQLite/MySQL – Database.
- HTML/CSS/JS – Web interface.

6 SPECIFICATION REQUIREMENTS

6.1 Interface Requirements

6.1.1 User Interface

Simple web portal for login and attendance; Teacher dashboard for records.

6.1.2 Hardware Interface

Device with camera (smartphone/laptop), classroom Wi-Fi.

6.1.3 Software Interface

Python + Flask backend, Database (MySQL/SQLite).

6.2 Functional Requirements

- Data Collection: Student identity and presence input.
- Data Processing: AI model validates identity + Wi-Fi connection.
- Attendance Generation: Securely logged in database.
- Output Delivery: Records shown in dashboards.

6.3 Non-Functional Requirements

- Performance: Fast < 3 sec.
- Scalability: Supports large classes.
- Reliability: Stable under load.

- Maintainability: Modular structure.
- Usability: Easy interface for users.

7 ALGORITHM

7.1 Attendance Marking Pipeline

1. Data Upload/User Input: Student initiates login.
2. Backend Processing: Flask verifies credentials.
3. Face Recognition & Wi-Fi Validation: AI confirms identity and network.
4. Token Verification: Validates session token.
5. Output Delivery: Attendance confirmed in database.

8 IMPLEMENTATION OF PROJECT

8.1 Data Collection

AI dataset of student faces created; student details stored in database.

8.2 Data Processing

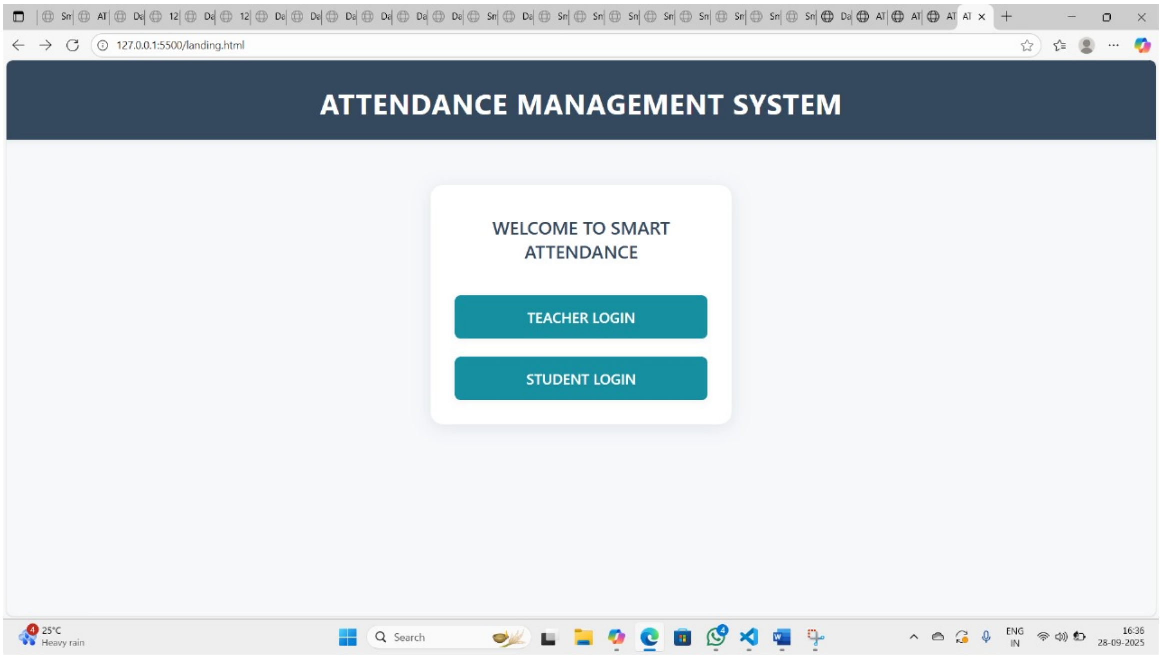
OpenCV face detection, Wi-Fi validation, token-based verification.

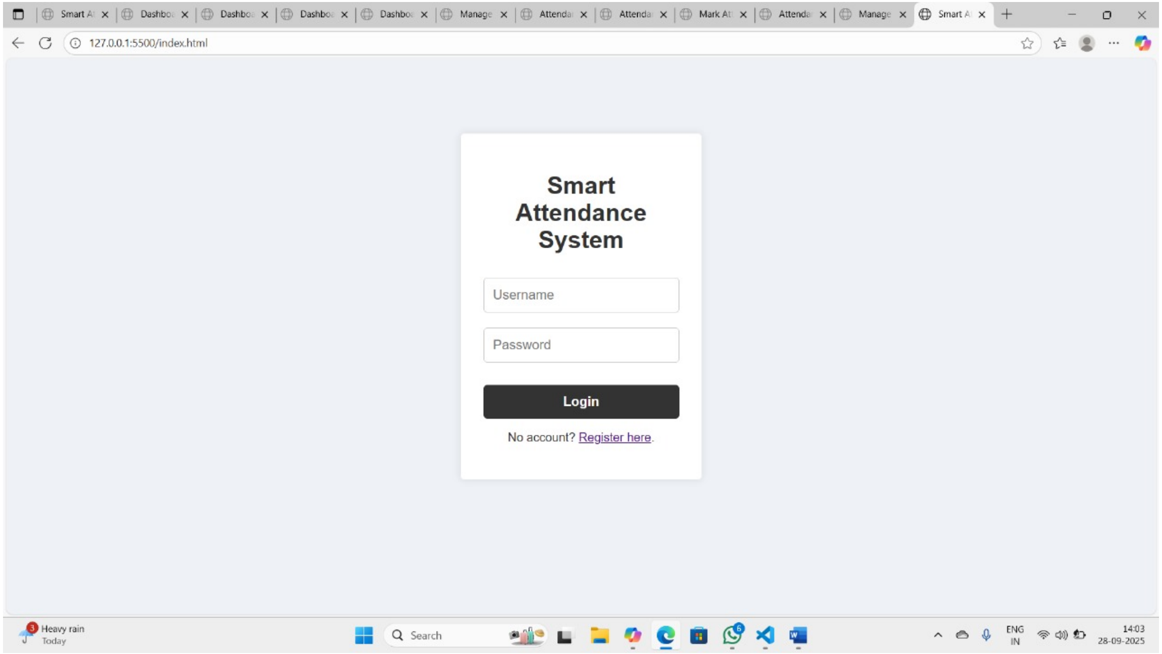
8.3 Visualization / Output Generation

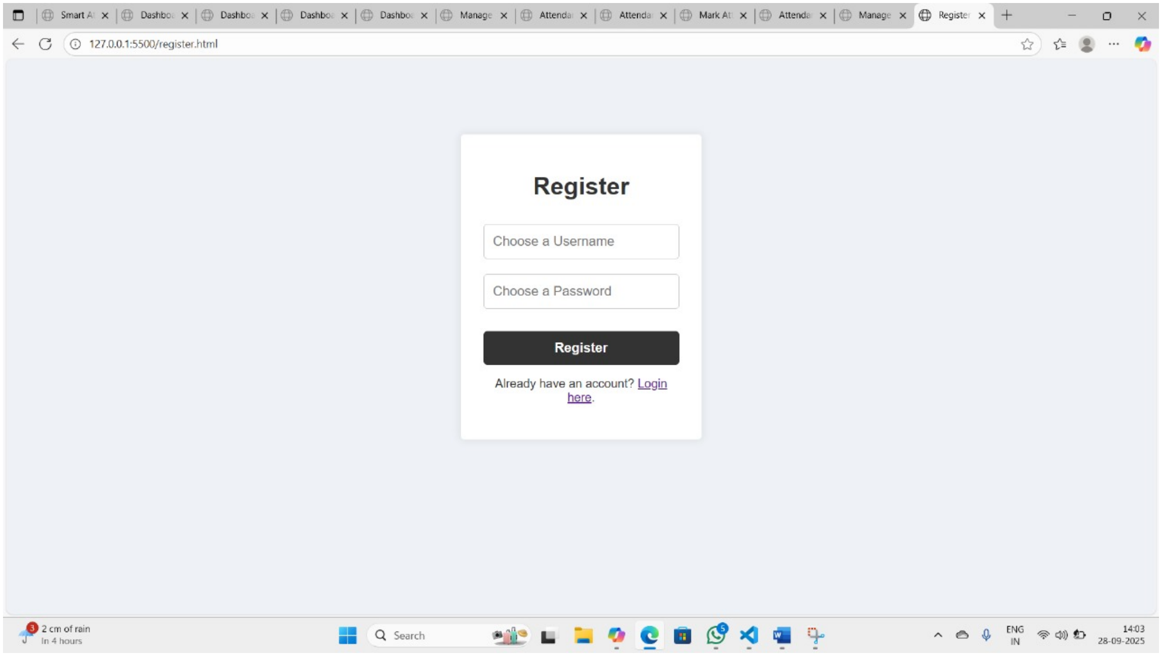
- Student Module: Attendance status view.
- Teacher Module: Manage and view class records.

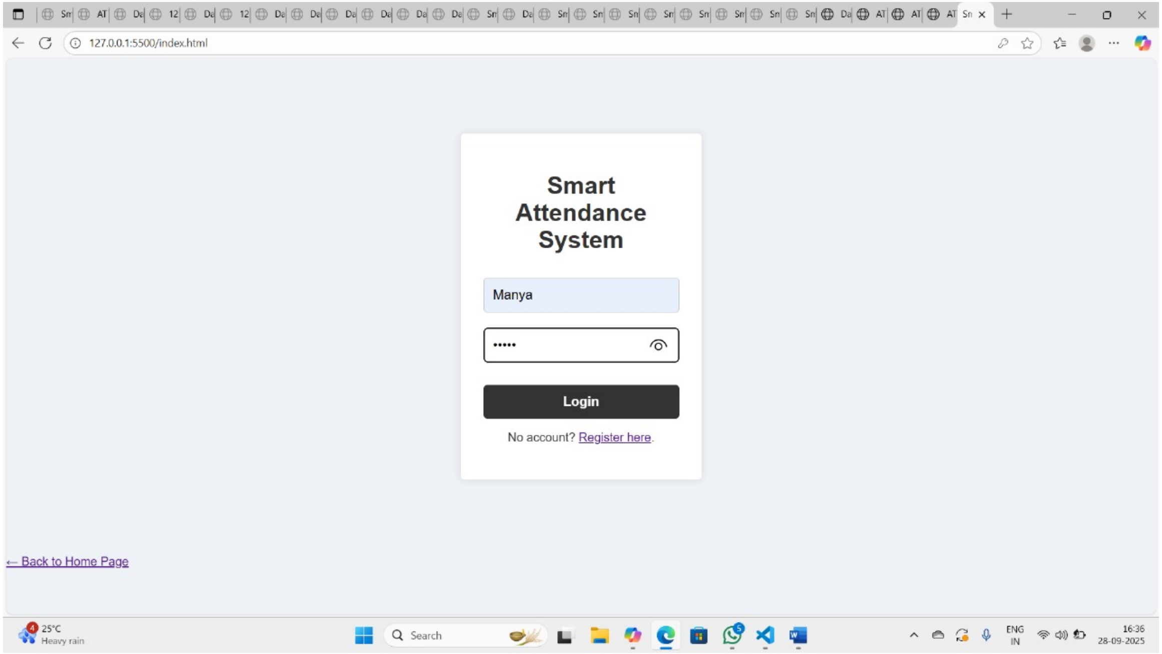
9 RESULT/OUTPUT

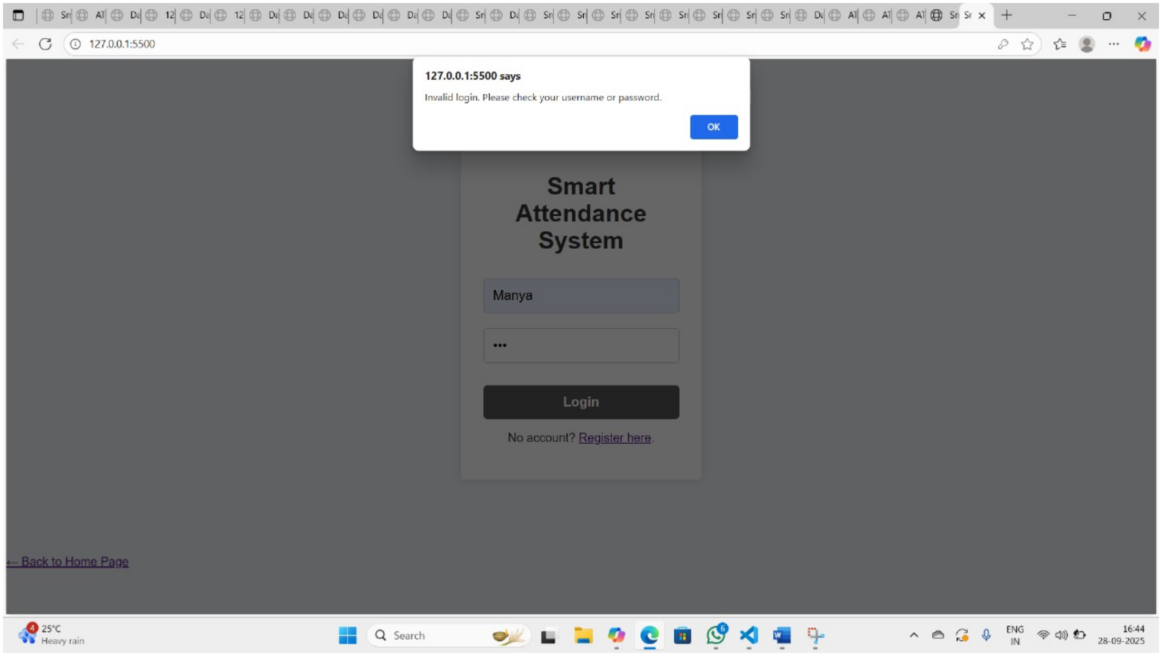
- Functional web application created.
- Login, Home, Attendance pages implemented.
- AI-powered recognition integrated.
- Attendance marked only after 3-step verification.
- Records stored in database, accessible via portal.

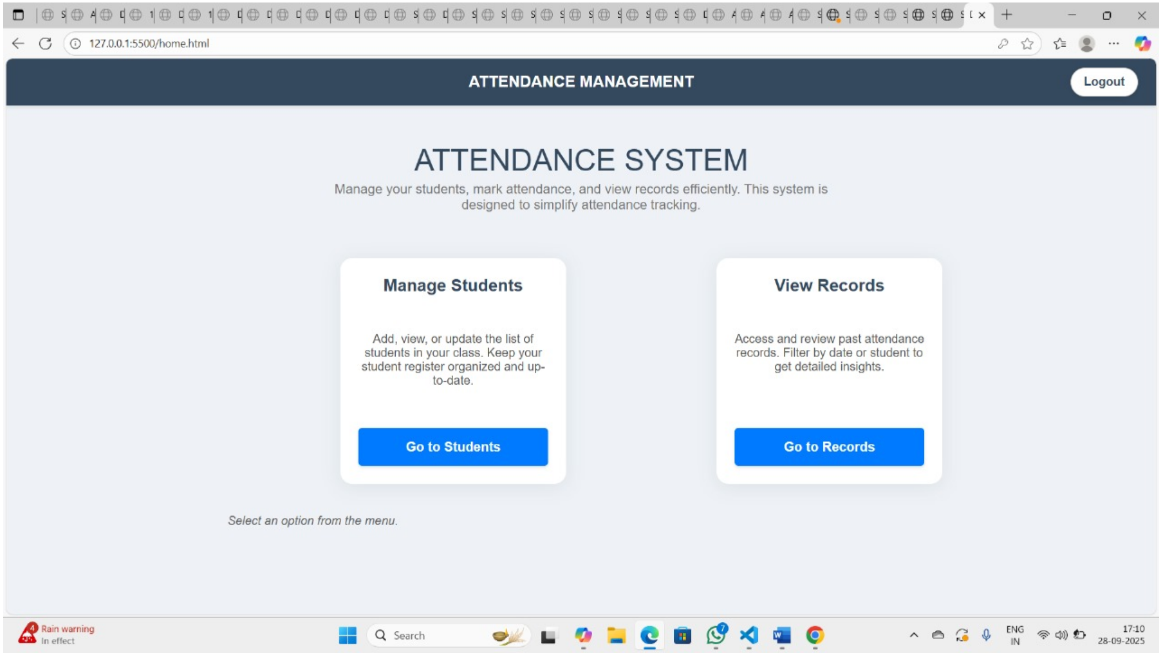


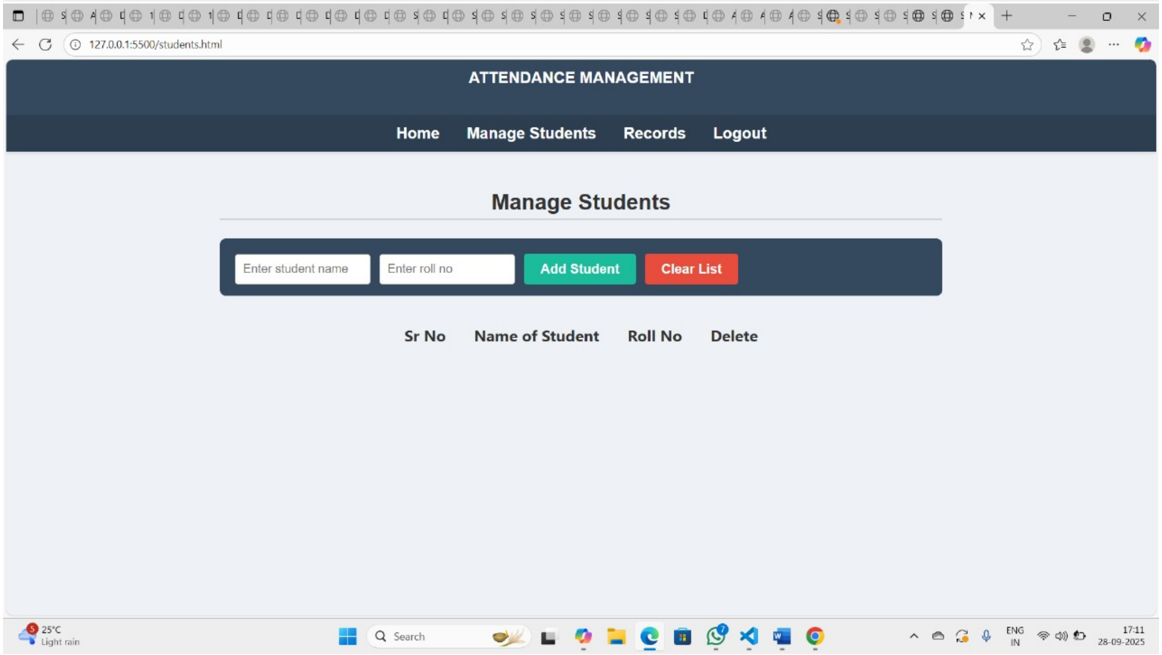

127.0.0.1:5500/index.html
Smart Attendance System
Username
Password
Login
No account? Register here.

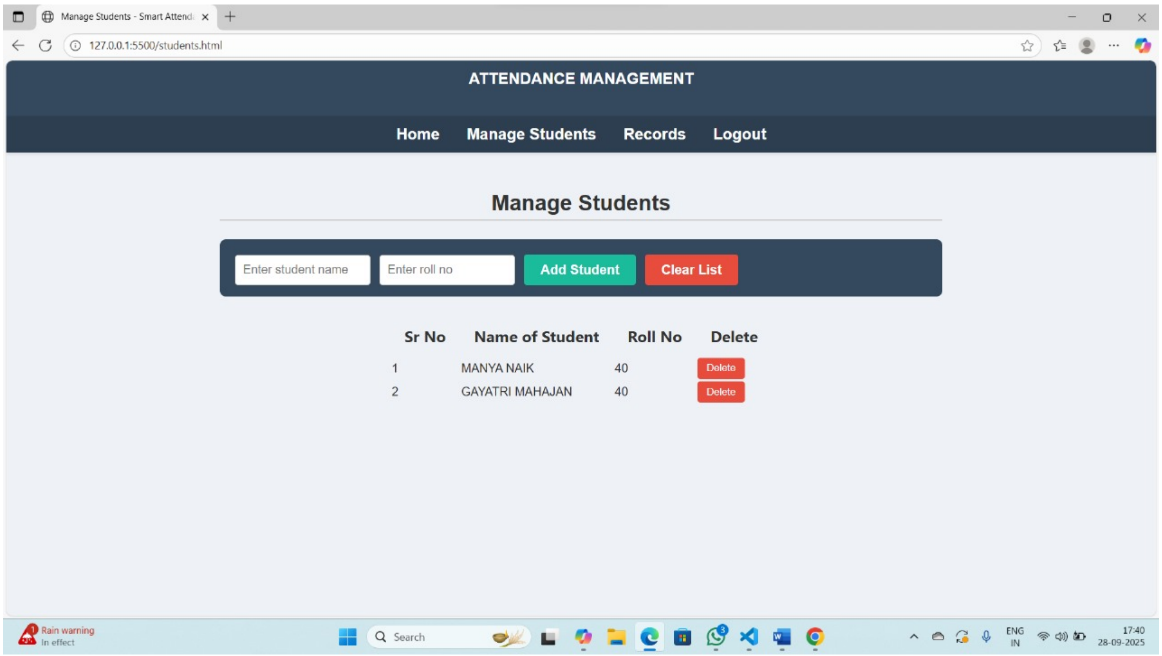

127.0.0.1:5500/register.html
Register
Choose a Username
Choose a Password
Register
Already have an account? Login here.

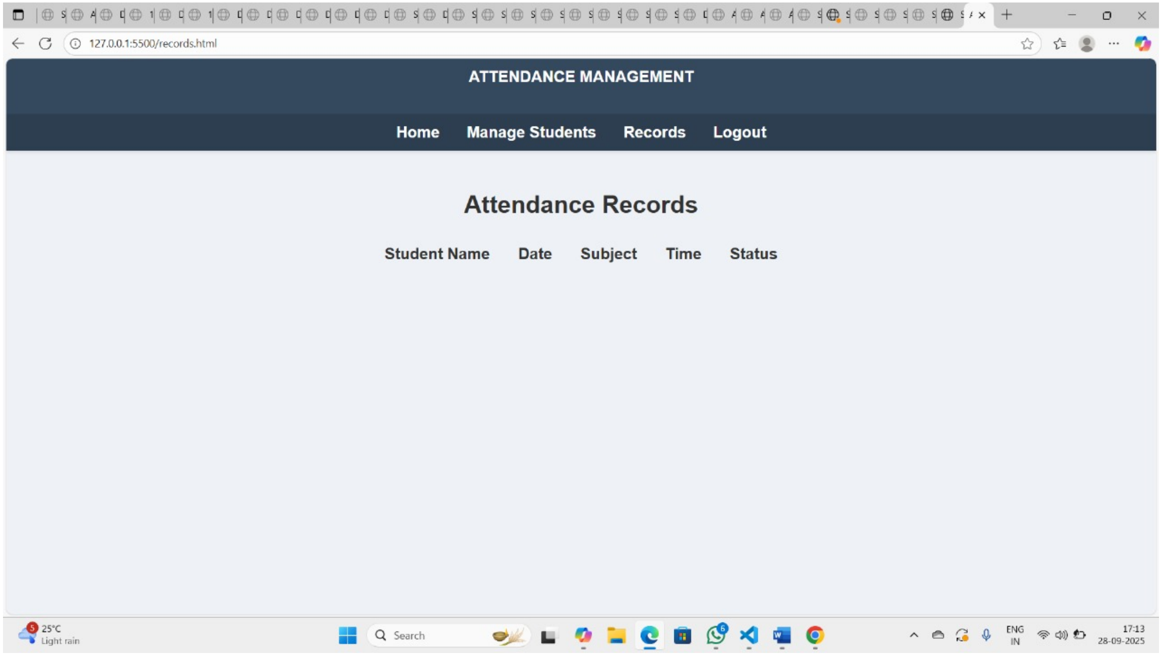

127.0.0.1:5500/index.html
Smart Attendance System
Manya
Login
No account? Register here.
← Back to Home Page


127.0.0.1:5500
127.0.0.1:5500 says
Invalid login. Please check your username or password.
OK
Smart Attendance System
Manya
Login
No account? Register here.
← Back to Home Page


127.0.0.1:5500/home.html
ATTENDANCE MANAGEMENT
Logout
ATTENDANCE SYSTEM
Manage your students, mark attendance, and view records efficiently. This system is designed to simplify attendance tracking.
Manage Students
Add, view, or update the list of students in your class. Keep your student register organized and up-to-date.
Go to Students
View Records
Access and review past attendance records. Filter by date or student to get detailed insights.
Go to Records
Select an option from the menu.


127.0.0.1:5500/students.html
ATTENDANCE MANAGEMENT
Home
Manage Students
Records
Logout
Manage Students
Enter student name
Enter roll no
Add Student
Clear List
Sr No
Name of Student
Roll No
Delete


Manage Students - Smart Attend
127.0.0.1:5500/students.html
ATTENDANCE MANAGEMENT
Home
Manage Students
Records
Logout
Manage Students
Enter student name
Enter roll no
Add Student
Clear List
Sr No
Name of Student
Roll No
Delete
1
MANYA NAIK
40
Delete
2
GAYATRI MAHAJAN
40
Delete
Rain warning
In effect
Search
ENG
IN
17:40
28-09-2025


127.0.0.1:5500/records.html
ATTENDANCE MANAGEMENT
Home
Manage Students
Records
Logout
Attendance Records
Student Name
Date
Subject
Time
Status
25°C
Light rain
Search
ENG
IN
17:13
28-09-2025

10 SCOPE OF THE SYSTEM

10.1 Project Scope

Automates attendance in classrooms, secure verification with AI, maintains reliable digital records.

10.2 Future Scope

- Mobile app with only essential attendance features.
- Low-cost devices provided to teachers for classroom use.
- Automatic face recognition for accurate student identification.
- Centralized database for secure attendance storage.
- Students can check only their own attendance.
- Eliminates proxy marking and reduces manual effort.
- Scalable to multiple classes, departments, or colleges.
- Potential for integration with academic systems and commercialization.

11 CONCLUSION AND REFERENCES

11.1 Conclusion

The **AI-Powered Smart Attendance Web Application** provides a modern, secure, and automated solution for attendance management. With **AI-based face recognition**, **Wi-Fi validation**, and **token verification**, it eliminates proxy, ensures fairness, and enhances academic integrity.

11.2 References

1. Smitha et al., "Automated Attendance Systems: Challenges and Opportunities," IJCS, 2020.

2. Ruhina B. Karani et al., "Image Registration using DCT," ICT Workshop, 2012.

3. Jimmy Singla and Raman Goyal, "Affine Transformation in Image Registration," IJITKM, 2004.

4. OpenCV Documentation – AI-based Face Recognition.

5. Flask Documentation – Web Framework for Python.

6. SQLite Documentation – Database Integration.